

Hotel Booking Website

The **Hotel Booking Website** is a Java Full Stack web application designed to provide users with a secure and user-friendly platform for hotel reservations. The system allows customers to browse hotels, search available rooms, apply promotional offers, and perform bookings online.

The project also includes an **Admin Portal** for centralized system and hotel management.

Technologies Used

Frontend Technologies

Technology	Purpose
ReactJS	UI Development
Vite	Fast frontend build tool
Tailwind CSS	Responsive styling
Axios	API communication
React Router DOM	Navigation and routing
Context API	State management
JWT Storage	Authentication

Backend Technologies

Technology	Purpose
Java	Core programming language
Spring Boot	Backend framework
Spring Security	Authentication & authorization
JWT	Token-based security
Spring Data JPA	Database operations
Hibernate	ORM mapping

Maven	Dependency management
Swagger	API documentation

Database Technologies

Technology	Purpose
MySQL	Relational database
JPA/Hibernate	Database mapping

Frontend Design

The frontend is designed using a **component-based architecture** for reusability and maintainability.

Main pages

Page	Description
Home.jsx	Displays hotel listings
BookingHistory.jsx	Shows booking history
CentralizedPortal.jsx	Admin management portal
ReceptionPortal.jsx	Receptionist dashboard
Login/Register	User authentication pages

Components

Component	Purpose
Navbar.jsx	Navigation bar
HotelList.jsx	Hotel display cards
SearchAndFilter.jsx	Search/filter functionality

BookingForm.jsx	Booking form
ReceptionistDashboard.jsx	Reception operations

Frontend Features

Customer Features

- Hotel search
- Room filtering
- Online booking
- Booking history
- Quick rebooking
- Promo code application

Admin Features

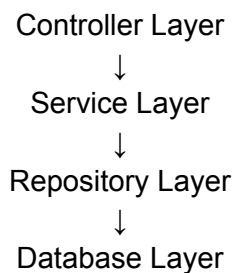
- Hotel management
- Room management
- Promotion management
- Receptionist management

Receptionist Features

- Daily arrivals/departures
- Check-in/check-out processing
- Booking modification
- Room assignment

Backend Design

The backend follows a layered architecture:



Controllers:

Controller	Purpose
AuthController	Login/Register APIs
HotelController	Hotel APIs
RoomController	Room APIs
BookingController	Booking APIs
ReceptionController	Reception operations

Services:

Service	Purpose
HotelService	Hotel management
BookingService	Booking logic
ReceptionService	Front desk operations
EmailService	Email notifications

Repositories:

Repository	Purpose
HotelRepository	Hotel database operations
RoomRepository	Room operations
BookingRepository	Booking operations

Security Layer

Security implementation includes:

- JWT authentication
- Spring Security
- Role-based authorization
- API rate limiting

Exception Handling

Handled exceptions include:

- RoomUnavailableException
- InvalidDateRangeException
- RateLimitExceededException
- UnauthorizedAccessException
- InvalidPromoCodeException
- UserNotFoundException
- InvalidBookingStatusTransitionException

Database Design

Role Enum

CUSTOMER, ADMIN, RECEPTIONIST

BookingStatus Enum

PENDING, CONFIRMED, CANCELLED, CHECKED_IN, CHECKED_OUT

User Table

Column	Type
id	Long
name	String
email	String
passwordHash	String
role	Enum

Hotel table

Column	Type
id	Long
name	String
location	String
facilities	String

Room Table

Column	Type
id	Long
hotelId	Long
roomCategory	String
price	Double
amenities	String
isAvailable	Boolean

Booking Table

Column	Type
id	Long
userId	Long
roomId	Long
reservationNumber	String
checkInDate	Date
checkOutDate	Date
totalAmount	Double
status	Enum

Promotion Table

Column	Type
id	Long

code	String
------	--------

discountAmount	Double
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type	String
------	--------

expiryDate	Date
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Entity Relationships

User → Booking (One-to-Many)

Hotel → Room (One-to-Many)

Room → Booking (One-to-Many)

Promotion → Booking (One-to-Many)

API URI Design

Authentication APIs

Method	URI	Description
POST	/api/auth/register	Register user
POST	/api/auth/login	Login user

Hotel APIs

Method	URI	Description
GET	/api/customer/hotels	Get all hotels
GET	/api/customer/hotels/{hotelId}/rooms	Get hotel details
POST	/api/admin/hotels	Add hotel (Admin)
GET	/api/customer/hotels/search	Available rooms filtered and retrieved based on search specifications

Room APIs

Method	URI	Description
GET	/api/customer/rooms	Get all rooms
GET	/api/customer/rooms/{id}	Get a particular room
POST	/api/admin/rooms	Add room
PUT	/api/admin/rooms/{id}/availability	Update room availability status

Booking APIs

Method	URI	Description
POST	/api/customer/bookings	Create booking
GET	/api/customer/bookings/history	Booking history
PUT	/api/customer/bookings/{bookingId}/rebook	Modify booking
GET	/api/customer/bookings/promotions/validate	Validate promo code

Promotion APIs

Method	URI	Description
GET	/api/admin/promotions	Get all promotions
POST	/api/admin/promotions	Add Promotions
GET	/api/customer/promotions/validate	Validate the customer promo code

Reception APIs

Method	URI	Description
GET	/api/reception/arrivals	Get all the Check-in customers for that certain date
GET	/api/reception/departures	Get all the Checked-out customers for that certain date
GET	/api/reception/occupancy	Current occupancy of the hotel
PUT	/api/reception/{bookingId}/check-in	Update customers Check-in information

PUT	/api/reception/{bookingId}/check-out	Update the customers Check-out information
PUT	/api/reception/{bookingId}/cancel	Cancels the booking
PUT	/api/reception/{bookingId}/assign-room	Physical rooms are assigned

Admin APIs

Method	URI	Description
GET	/api/admin/bookings	Retrieve all bookings for administrative overview
POST	/api/admin/receptionists	Create a new receptionist account

Business Rules

Rule	Description
Date Validation	Check-in must be before check-out
Availability Update	Room availability updates instantly
Rate Limiting	Secure APIs are rate limited
Authentication Required	Booking history requires login
Promo Validation	Expired promotions are invalid
Reception Access	Only Admin/Receptionist can check-in/check-out

Actions Based on Roles

Customer Actions

- Register/Login
- Search hotels
- Filter rooms
- Book hotel rooms
- View booking history
- Apply promo codes

- Rebook previous reservations

Admin Actions

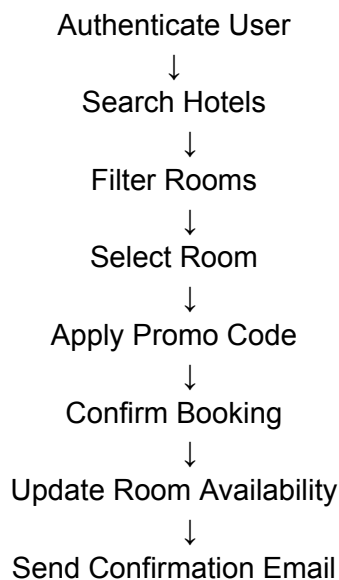
- Manage hotels
- Manage room categories
- Monitor bookings
- Create promotional offers
- Manage receptionist accounts

Receptionist Actions

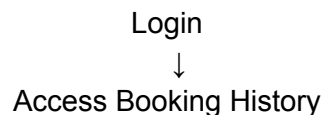
- View arrivals/departures
- Process customer check-ins
- Process customer check-outs
- Modify bookings
- Assign physical rooms

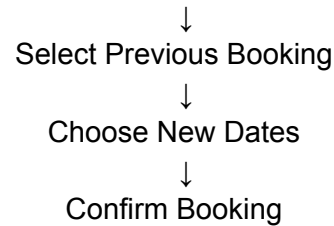
Workflows

Workflow 1 – New Booking

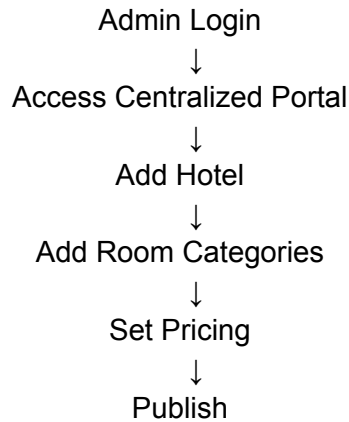


Workflow 2 – Quick Rebooking

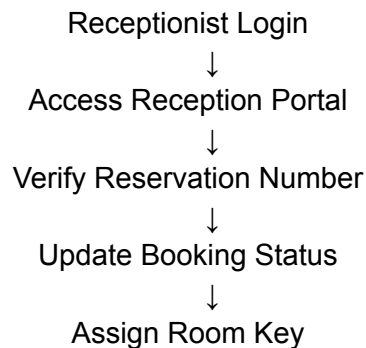




Workflow 3 – Admin Setup



Workflow 4 – Front Desk Operations



Conclusion

The Hotel Booking Website is a complete Java Full Stack application that provides secure and efficient hotel reservation services for customers while offering centralized management capabilities for administrators.

The system combines:

- Modern frontend development using ReactJS
- Secure Spring Boot REST APIs
- JWT authentication
- MySQL database management
- Scalable layered architecture

This project demonstrates important real-world software engineering concepts including:

- Role-based authentication
- RESTful API development
- Database relationships
- Full-stack integration
- Business rule validation
- Secure booking workflows

The application can further be enhanced with:

- Online payment integration
- Email/SMS notifications
- Advanced analytics dashboard
- Real-time chat support
- Cloud deployment